2. AMPLITUDE MODULATION

This experiment comprises two parts, the simulation and the hardware. The simulation part uses three different circuit configurations for demonstration of amplitude modulation, which is useful in telephony and audio broadcasting. The hardware part uses one circuit configuration. The other configurations can also be built using hardware.

2.1. OBJECTIVES

a) Demonstrate an amplitude-modulated carrier in the time and frequency domains for different modulation indices and modulating frequencies.

b) Determine the modulation index, percent modulation, signal bandwidth and side frequencies of amplitude-modulated carrier.

2.2. SIMULATION/HARDWARE

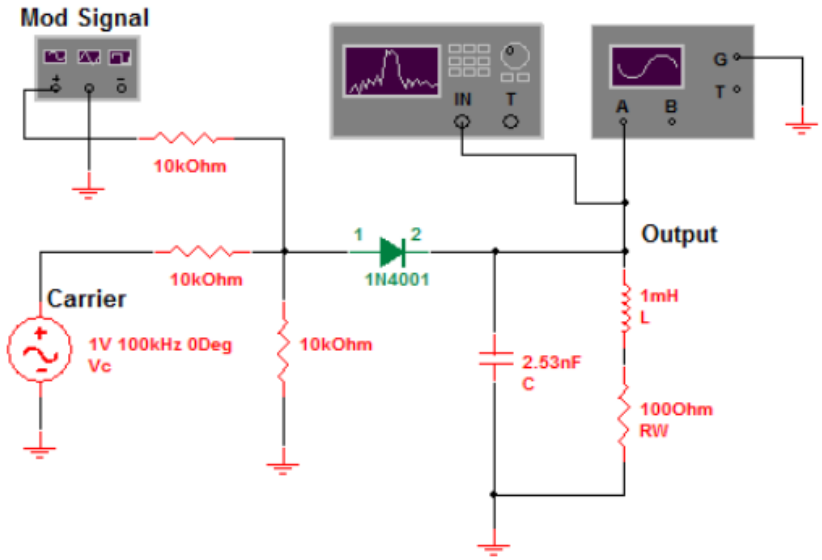
This experiment will first use the Electronics Workbench (Multisim) simulation followed by hardware. ***Please remember to label your plots appropriately on the time and amplitude axes or frequency and amplitude axes.***

2.2.1. SIMULATION

This portion of the experiment will use the Electronics Workbench (Multisim) simulation.

2.2.1.1. PROCEDURE

1. **<u>Diode Amplitude Modulator:</u>** Set up the circuit of the diode amplitude modulator as illustrated in figure below. The modulating signal (sine wave) frequency should be 5 kHz, with amplitude 1 V. Set the carrier frequency to 100 kHz and amplitude to 1 V. Display an amplitude-modulated carrier on the scope and draw the plot. Also calculate the modulation index and percent modulation based on baseband (V_m) and carrier (V_c) amplitudes. Compare them with the ones determined from the plot.



2. Display the frequency spectrum of the modulated carrier on the spectrum analyser. Draw the spectral plot. Measure the carrier frequency (f_c), the upper side frequency (f_{usf}), the lower side frequency (f_{lsf}), and the voltage amplitude of each spectral line and show them on the spectral plot. Determine the bandwidth from the plot.

3. Change the modulating signal amplitude to 0.5 V and repeat step 1 above. How did the values of the modulation index and percent modulation compare with the initial ones?

4. Display the frequency spectrum of the new modulated carrier on the spectrum analyser. How do the side frequencies' amplitudes and the bandwidth from this frequency spectrum compare with the ones in step 2 above?

5. Change the modulating signal amplitude to 0 V and repeat step 1 above. How did the wave shape compare with the previous amplitude-modulated waveforms? Explain any difference.

6. Draw the new spectral plot and compare it with the previous ones.

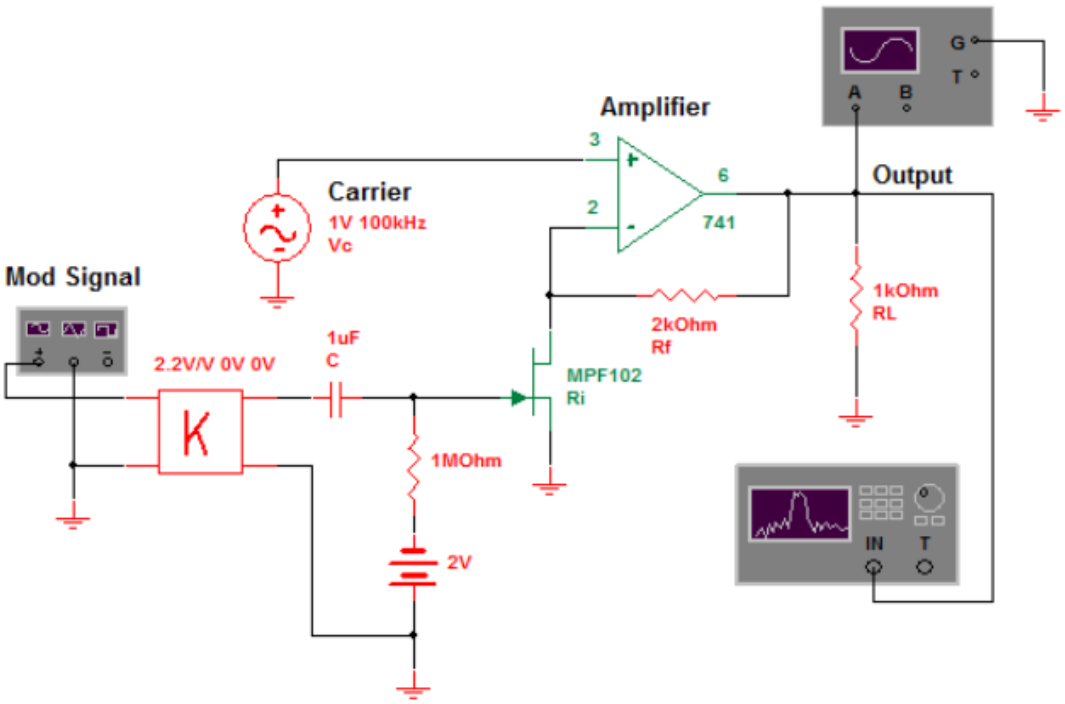
7. Change the modulating frequency to 10 kHz and the amplitude back to 1 V. Again display the amplitude-modulated carrier on the scope and draw the plot. How does the waveform differ from the waveform for a 5 kHz modulating frequency in step 1?

8. Connect the spectrum analyser and repeat step 2 for the 10 kHz modulating signal. How does the bandwidth of the modulated carrier for a 10 kHz modulating frequency compare with the bandwidth for a 5 kHz modulating frequency? Was the any difference between the amplitude of the side frequencies for the 10 kHz plot and the 5 kHz plot? Explain.

9. Change the modulating signal frequency back to 5 kHz and select square wave on the function generator. Display the carrier modulated by a square wave on the oscilloscope screen. Draw the plot. How does the wave shape differ from the wave shape in step 1? Determine the modulation index and percent modulation from the plot.

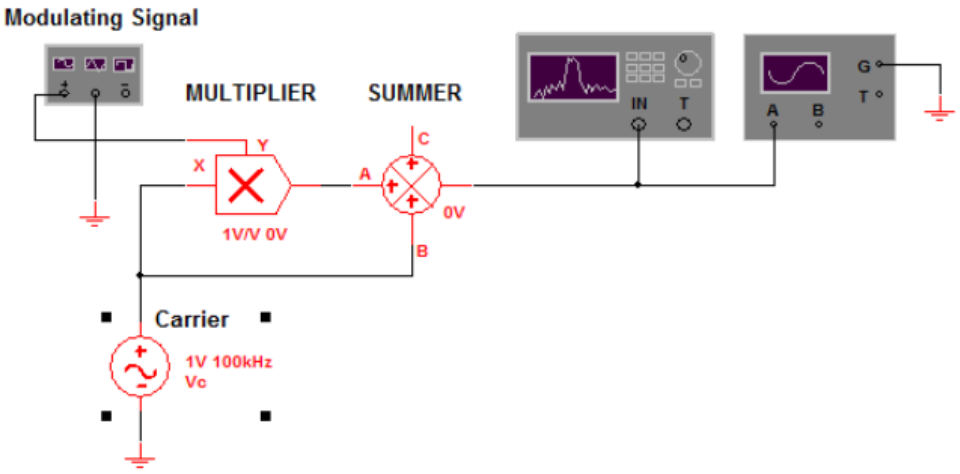
10. Use the spectrum analyser to display the spectrum for the square wave modulated carrier. Draw the spectral plot and ignore any side frequencies with amplitudes less than 10% of the carrier amplitude. Measure the frequency of the spectral lines and record answers on the plot. How does the frequency spectrum for the square-wave modulated carrier differ from the spectrum for the sine wave modulated carrier? Explain why they are different. Determine the bandwidth of the modulated from the spectral plot and compare it with the bandwidth for the 5-kHz sine wave modulated carrier?

11. **AM using op-amp, transistor and voltage gain block**: Set up the circuit of an amplitude modulator using op-amp, transistor and voltage gain block as shown in the figure below. The modulating signal (sine wave) frequency should be 5 kHz, with amplitude V_m = 1 V. Set the carrier frequency to 100 kHz and amplitude to V_c = 1 V. Display an amplitude-modulated carrier on the scope and draw the plot. Also calculate the modulation index and percent modulation based on baseband (V_m) and carrier (V_c) amplitudes. Compare them with the ones determined from the plot.



12. Display the frequency spectrum of the modulated carrier on the spectrum analyzer. Draw the spectral plot. Measure the carrier frequency (fc), the upper and lower side frequencies, and the voltage amplitude of each spectral line and show them on the spectral plot. Determine the bandwidth from the plot.

13. Set up the circuit of an amplitude modulator using multiplier and voltage summer as shown in the figure below. The modulating sine wave frequency should be 5 kHz, with amplitude V_m = 1 V. Set the carrier frequency to 100 kHz and amplitude to V_c = 1 V. Display an amplitude-modulated carrier on the scope and draw the plot. Also calculate the modulation index and percent modulation based on baseband (V_m) and carrier (V_c) amplitudes. Compare them with the ones determined from the plot.



14. Repeat step 12 for this circuit.

2.2.2 HARDWARE

The following **equipment** would be required:

One DC Voltage supply
Two Function Generators
One Spectrum analyser
One dual-trace oscilloscope
One 1N4001 Diode
Capacitor: 2.53 nF
Inductor: 1 mH
Resistors: 100, 2K, 10K, 20K

2.2.2.1 PROCEDURE

Repeat steps 1 to 10 as outlined in section 2.2.1.1